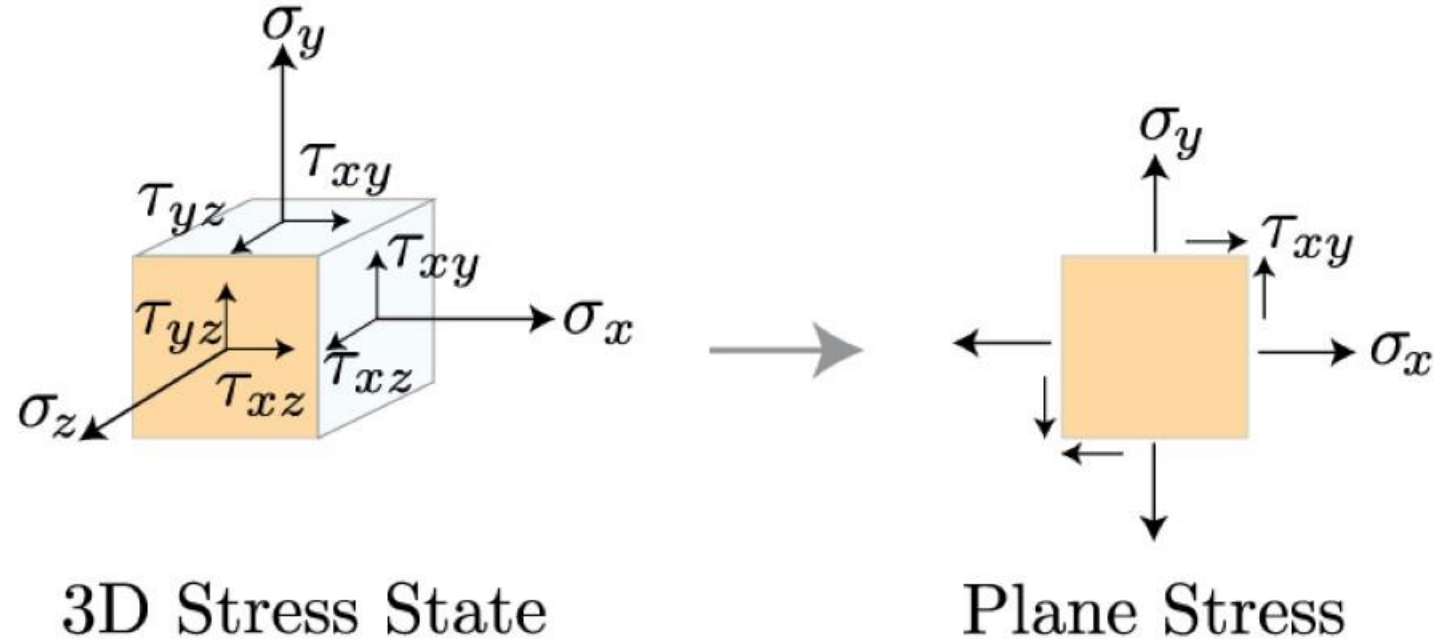
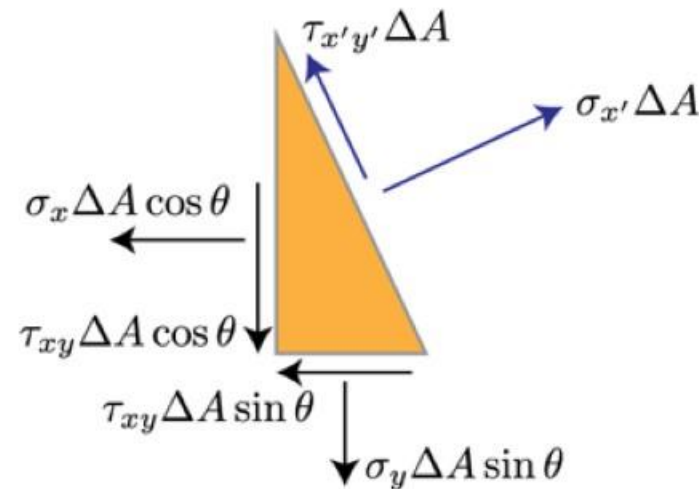
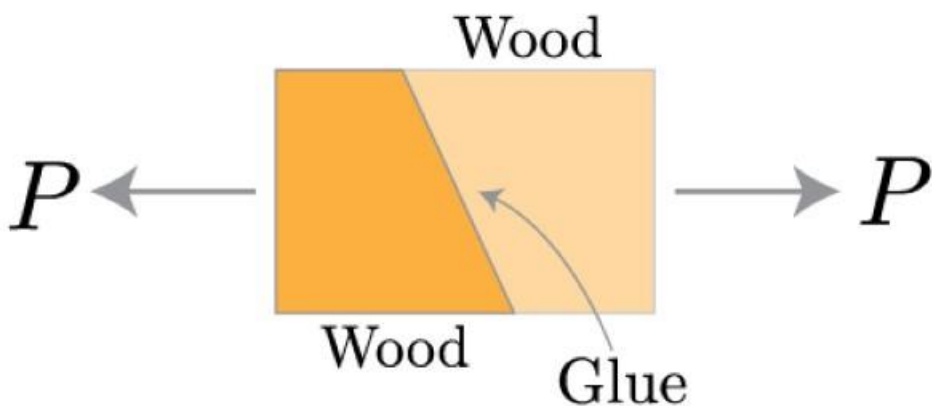




$$\sigma = \begin{pmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{pmatrix} \rightarrow \sigma = \begin{pmatrix} \sigma_x & \tau_{xy} & 0 \\ \tau_{xy} & \sigma_y & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Stress Transformation in 2D



$$\rightarrow \sum F_{x'} = 0$$

$$0 = \sigma_{x'} \Delta A - (\tau_{xy} \Delta A \sin \theta) \cos \theta - (\tau_{xy} \Delta A \cos \theta) \sin \theta - (\sigma_x \Delta A \cos \theta) \cos \theta - (\sigma_y \Delta A \sin \theta) \sin \theta$$

$$\sigma_{x'} = \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + \tau_{xy} (2 \sin \theta \cos \theta)$$

$$\uparrow \sum F_{y'} = 0$$

$$0 = \tau_{x'y'} \Delta A + (\tau_{xy} \Delta A \sin \theta) \sin \theta - (\tau_{xy} \Delta A \cos \theta) \cos \theta - (\sigma_y \Delta A \sin \theta) \cos \theta + (\sigma_x \Delta A \cos \theta) \sin \theta$$

$$\tau_{x'y'} = (\sigma_y - \sigma_x) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

Trigonometric Identities

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

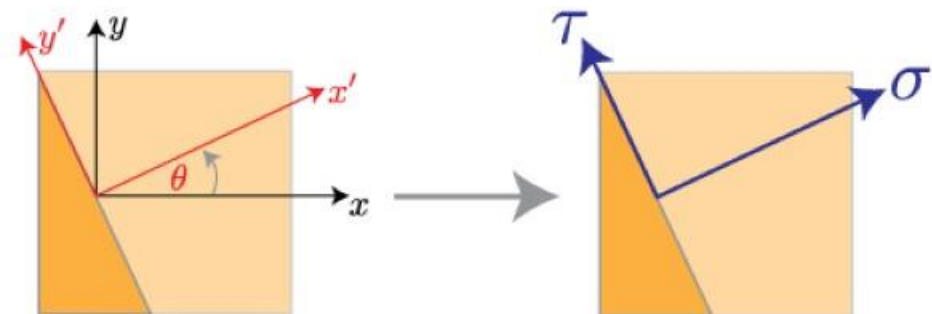
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\frac{d\sigma_{x'}}{d\theta} = 0$$

$$0 = -\frac{\sigma_x - \sigma_y}{2} (2 \sin 2\theta) + 2\tau_{xy} \cos 2\theta$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}}$$



Stress Transformation in 2D

$$\sigma_{x'x'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} + \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta - \tau_{xy} \sin 2\theta = \frac{\sigma_1 + \sigma_3}{2} + \frac{\sigma_1 - \sigma_3}{2} \cos 2\theta$$

$$\sigma_{y'y'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} - \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta - \tau_{xy} \sin 2\theta = \frac{\sigma_1 + \sigma_3}{2} - \frac{\sigma_1 - \sigma_3}{2} \cos 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\theta + \tau_{xy} \cos 2\theta = -\frac{\sigma_1 - \sigma_3}{2} \sin 2\theta$$

Principal Stresses: Principal Plane - Zero shear stress

$$\sigma_{1,2} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_{xx} - \sigma_{yy}}$$

$$\theta_{p2} = \theta_{p1} + 90^\circ$$

Maximum Shear Stresses

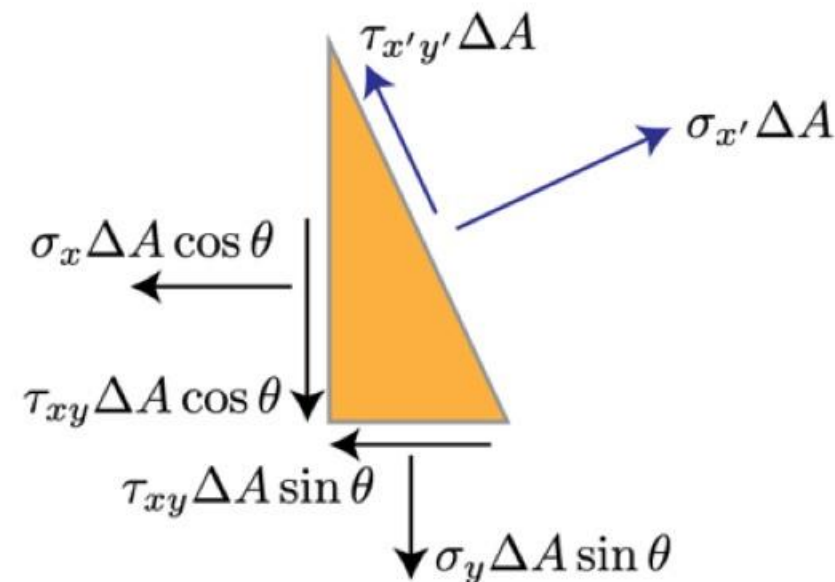
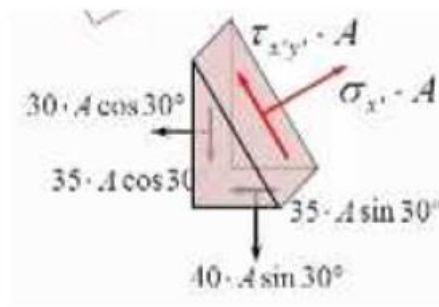
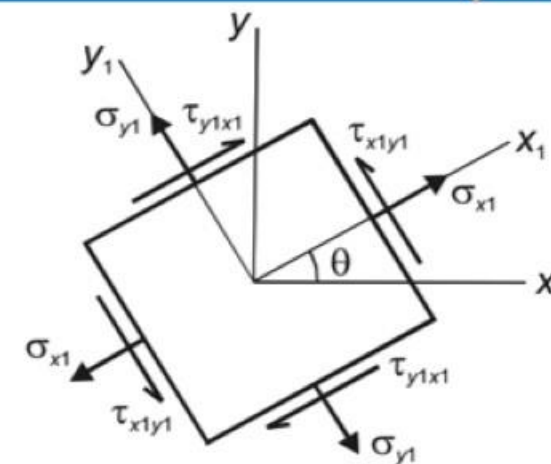
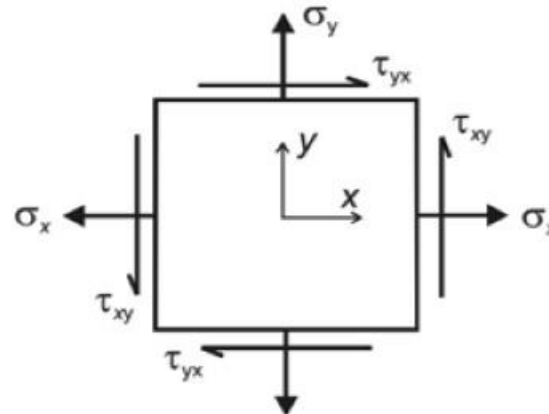
$$\tau_{max} = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2} = \frac{\sigma_1 - \sigma_3}{2}$$

$$\tan 2\theta_s = \frac{-(\sigma_{xx} - \sigma_{yy})}{2\tau_{xy}}$$

$$\theta_{s2} = \theta_{s1} + 90^\circ$$

Normal stress on maximum shear plane is

$$\sigma = \frac{\sigma_1 + \sigma_3}{2} = \frac{\sigma_{xx} + \sigma_{yy}}{2}$$



Mohr's Circle

✓ A graphical representation of normal and shear stresses on any plane inclined to the principal plane.

✓ Devised by a German scientist , Otto Mohr

✓ Centre of circle is $\left(\frac{\sigma_1 + \sigma_3}{2}, 0\right)$ or $\left(\frac{\sigma_{xx} + \sigma_{yy}}{2}, 0\right)$

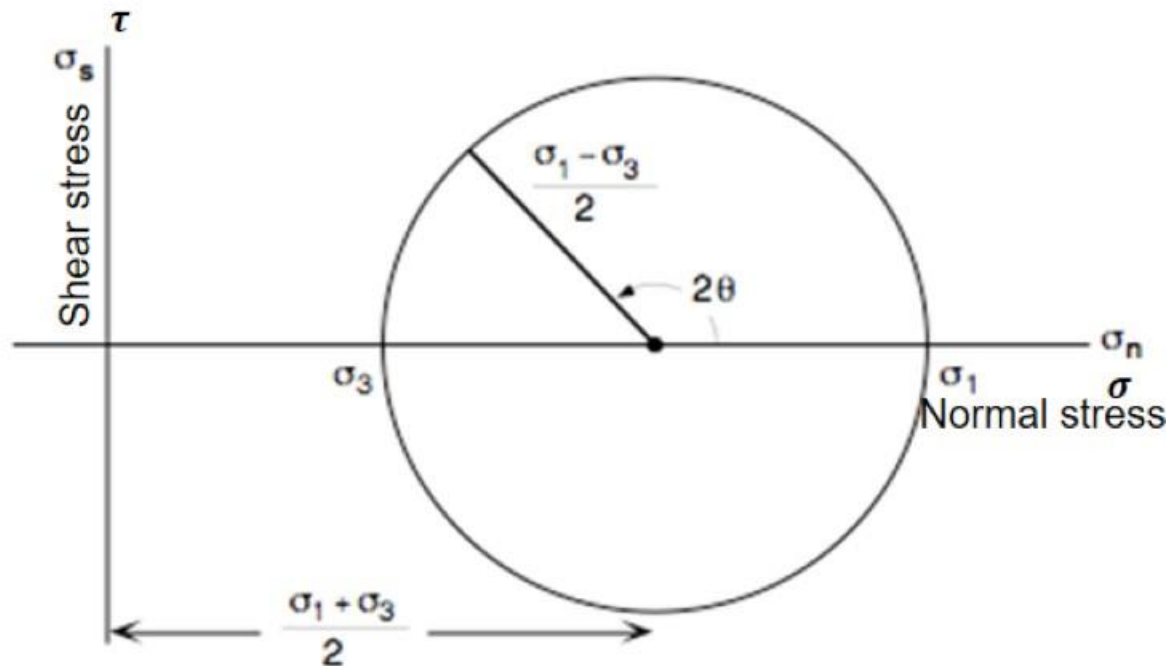
✓ Radius of circle is $\sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$ or, $\frac{\sigma_1 - \sigma_3}{2}$

✓ $\sigma_1 + \sigma_3 = \sigma_{xx} + \sigma_{yy}$

✓ $\sigma_{avg} = \frac{\sigma_{xx} + \sigma_{yy}}{2} = \frac{\sigma_1 + \sigma_3}{2}$

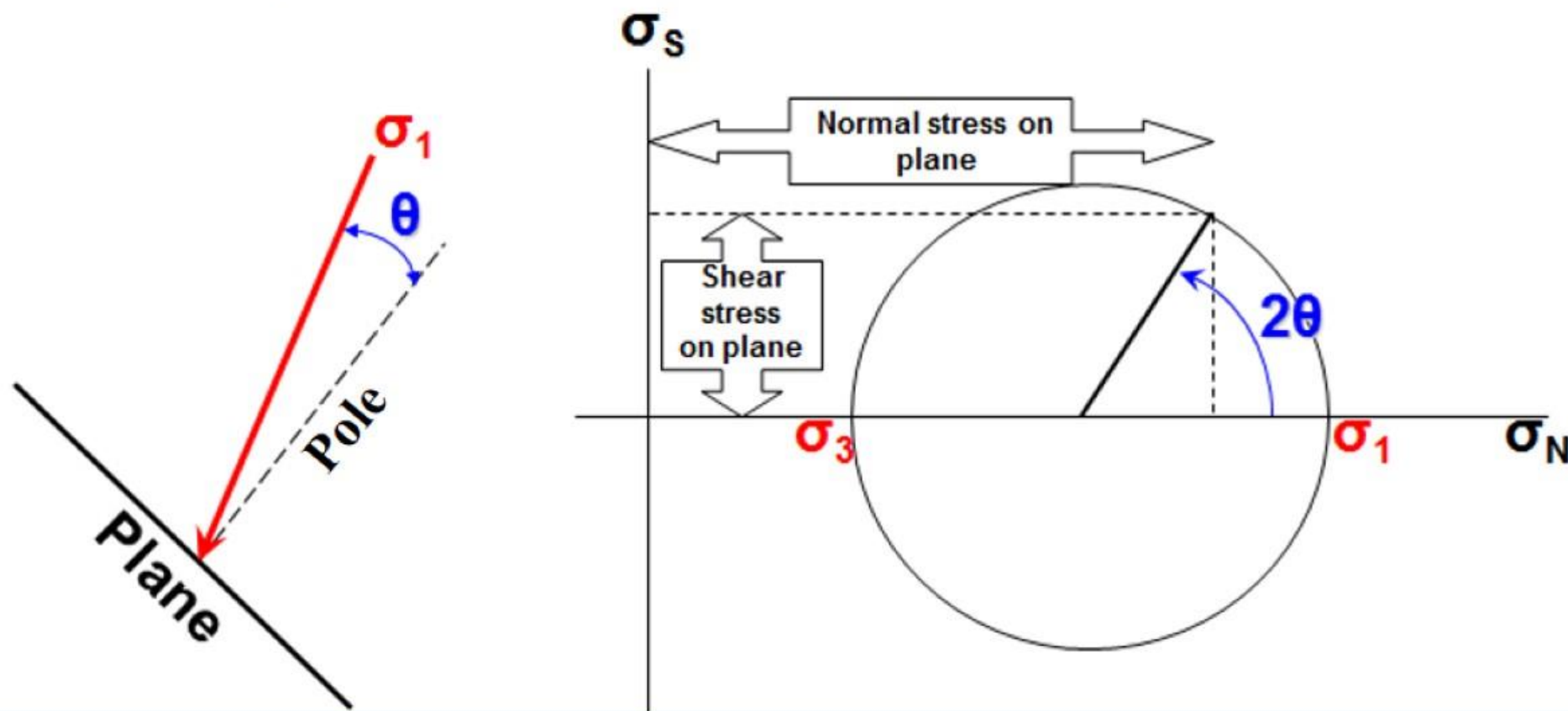
✓ $\left(\sigma - \frac{\sigma_{xx} + \sigma_{yy}}{2}\right)^2 + \tau^2 = \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2$

✓ $(\sigma - \sigma_{avg})^2 + \tau^2 = \left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2$



Mohr's Circle

- ✓ In Mohr's circle, Normal stresses - Abscissa and Shear stresses - Ordinate
- ✓ σ_θ and τ_θ are plotted for all possible values of θ , the locus is a circle.
- ✓ The centre of the circle - axis of normal stress.
- ✓ Centre : $\tau = 0$; $\sigma = \frac{\sigma_1 + \sigma_3}{2}$ and Radius = $\frac{\sigma_1 - \sigma_3}{2}$.





Characteristics of Mohr's Circle

- ✓ Maximum shear stress: $\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$;
- ✓ Maximum shear stress planes inclined at 45° to the principal plane.
- ✓ Normal stresses on maximum shear stress planes will be equal to each other: $\sigma = \frac{\sigma_1 + \sigma_3}{2}$
- ✓ Any point on the circle represents stresses on a plane which makes an angle θ with the major principal plane
- ✓ Resultant stress on that plane = $\sqrt{\sigma^2 + \tau^2}$
- ✓ Angle of obliquity, $\beta = \tan^{-1} \left(\frac{\tau}{\sigma} \right)$
- ✓ Maximum angle of obliquity - A Tangent to the circle from the origin. $\beta_m = \sin^{-1} \left(\frac{\sigma_1 - \sigma_3}{\sigma_1 + \sigma_3} \right)$
- ✓ Shear stress, τ_f on the plane of maximum obliquity is less than the maximum shear stress τ_{max} ($\tau_f < \tau_{max}$)
- ✓ Sum of normal stresses on mutually perpendicular planes is a constant $\sigma_\theta + \sigma_{\theta'} = \sigma_1 + \sigma_3$
- ✓ As Mohr circle is symmetrical about σ axis - only the top half circle is drawn.
- ✓ Stresses on conjugate planes are equal and opposite .
- ✓ Isotropic Point - Shear stresses vanish on all points.

Characteristics of Mohr's Circle

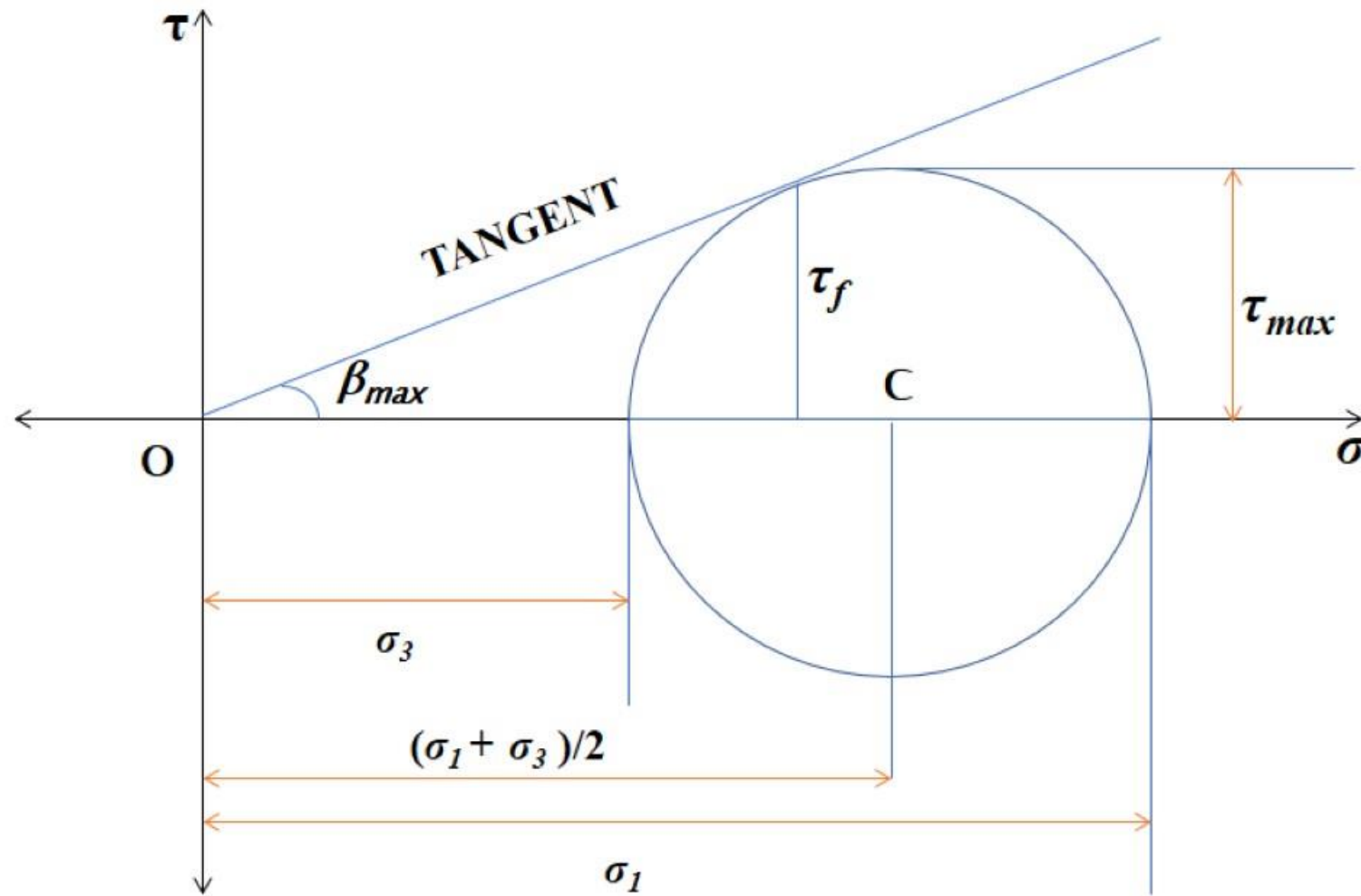
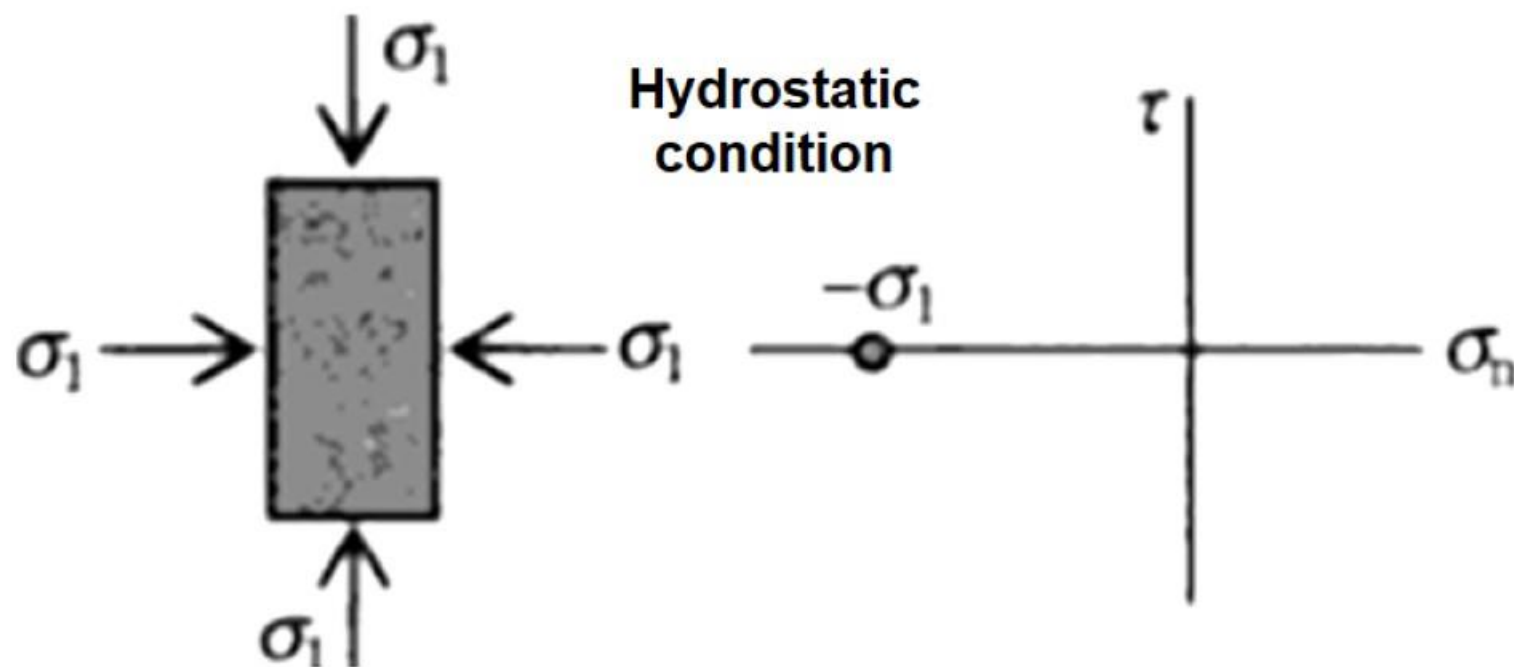


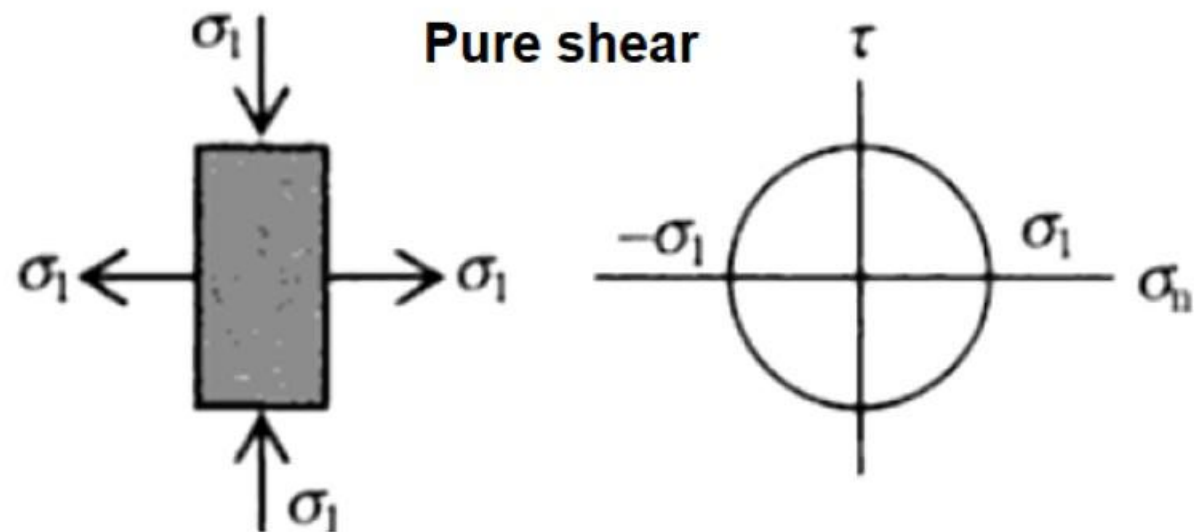
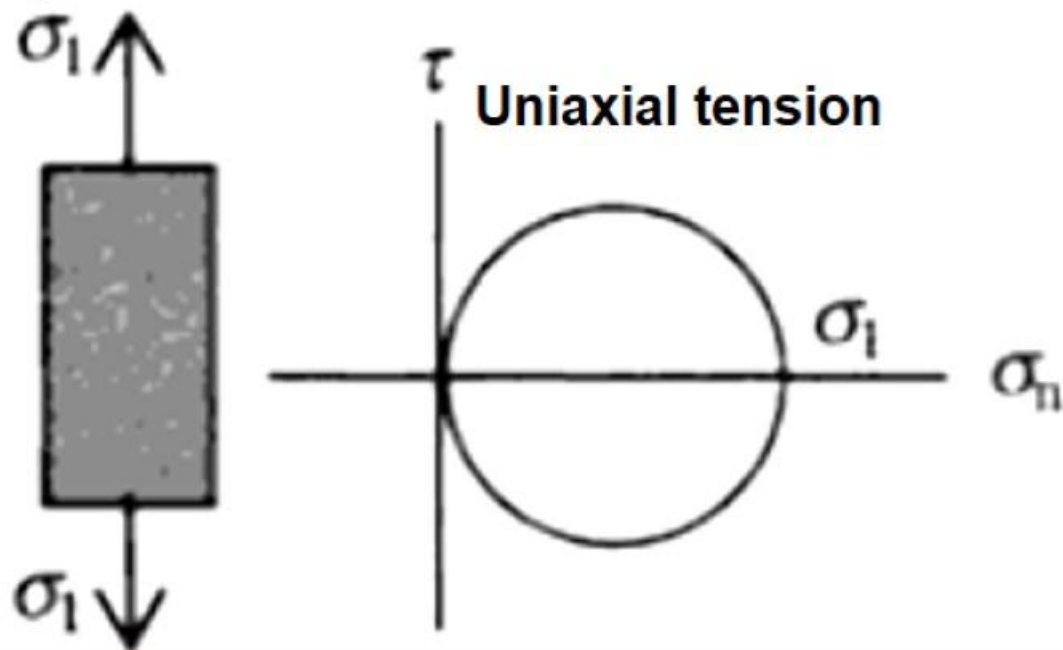
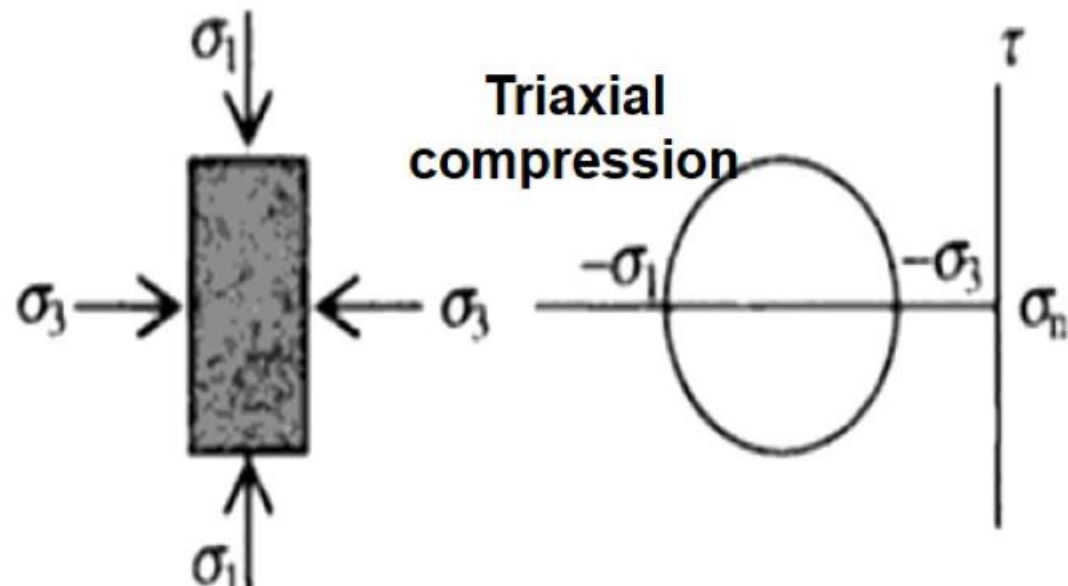
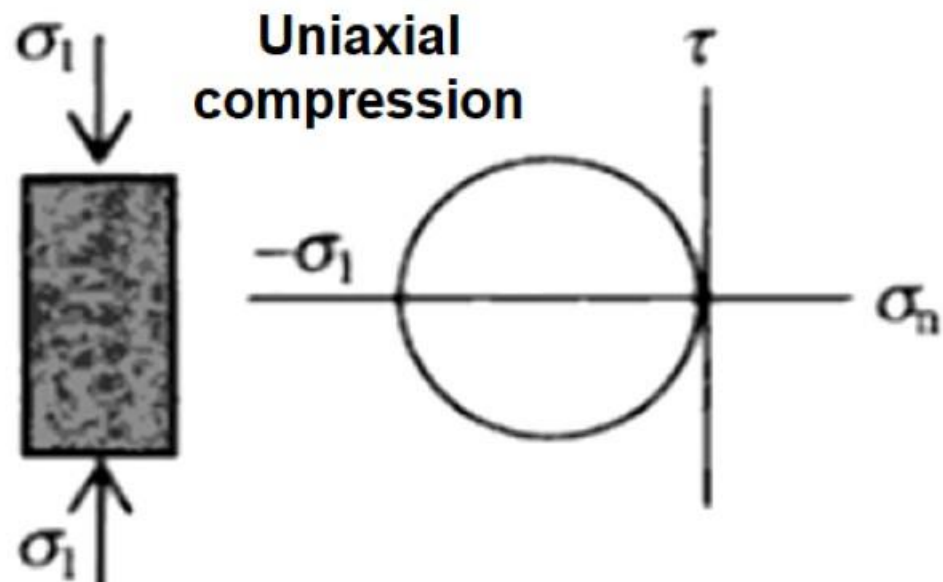
Fig: Mohr's Circle

Mohr's Circle for Special Stress Conditions

- ✓ Uniaxial compression
- ✓ Uniaxial tension
- ✓ Triaxial compression
- ✓ Pure shear
- ✓ Hydrostatic stress condition



Mohr's Circle for Special Stress Conditions



Mohr's Failure Criteria

- ✓ The Mohr-Coulomb failure criterion can be written as the equation for the line that represents the failure envelope. The general equation is

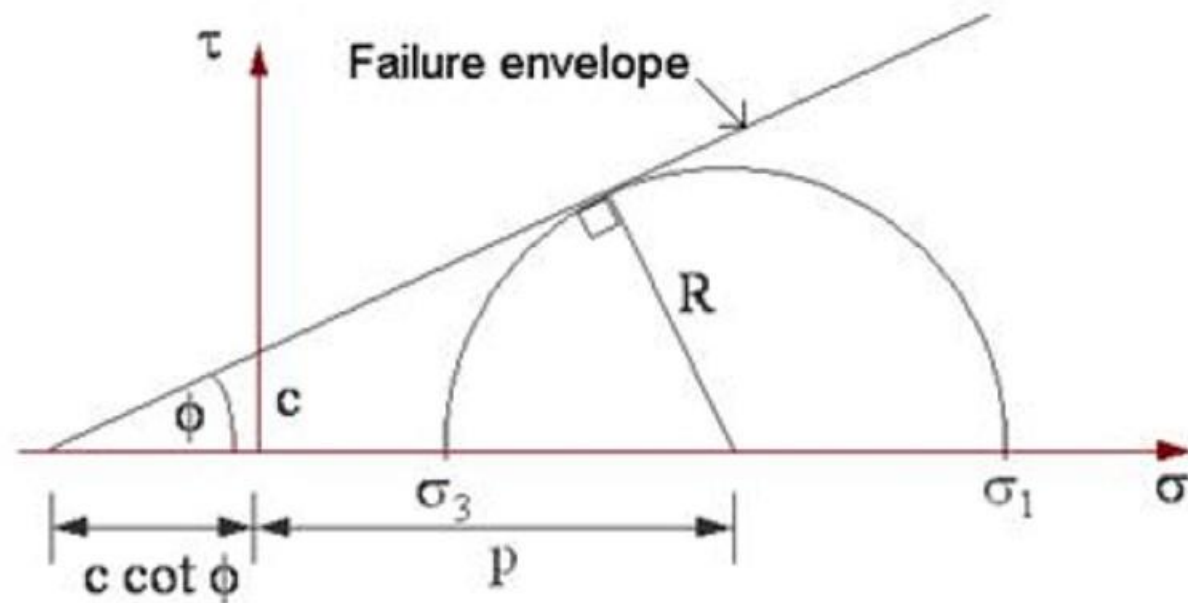
$$\tau_f = c + \sigma_f \tan \varphi$$

- ✓ Where τ_f = shear stress on the failure plane, c = apparent cohesion, σ_f = normal stress on the failure plane, φ = angle of internal friction. The failure criterion can be expressed in terms of the relationship between the principal stresses. From the geometry of the Mohr circle,

$$\sin \varphi = \frac{R}{c \cot \varphi + p} = \frac{\frac{\sigma_1 - \sigma_3}{2}}{c \cot \varphi + \frac{\sigma_1 + \sigma_3}{2}}$$

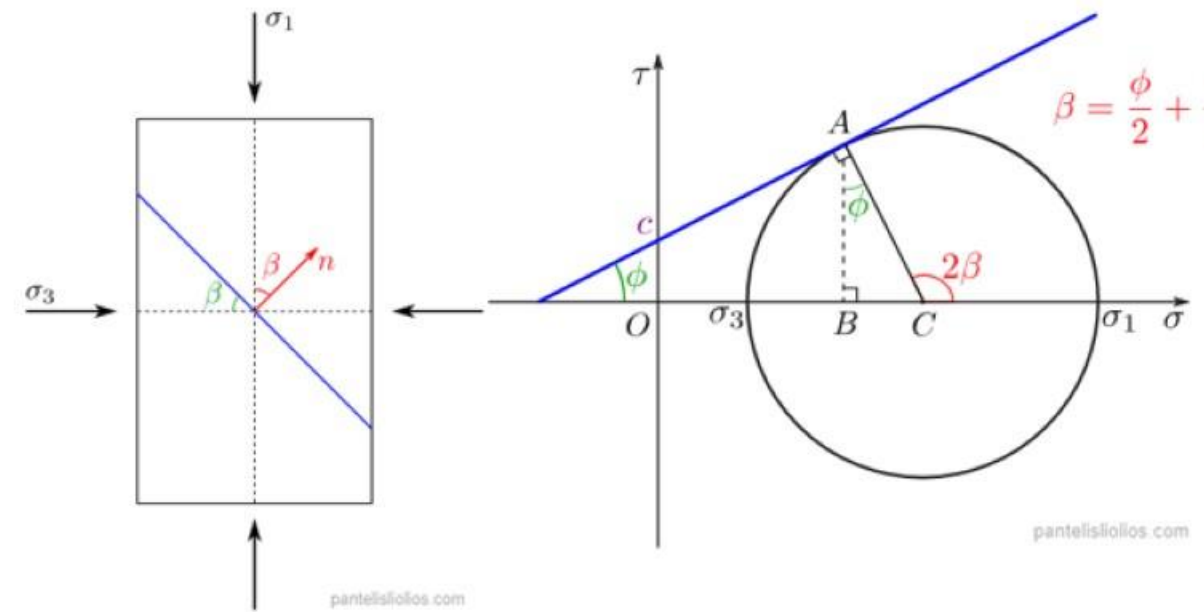
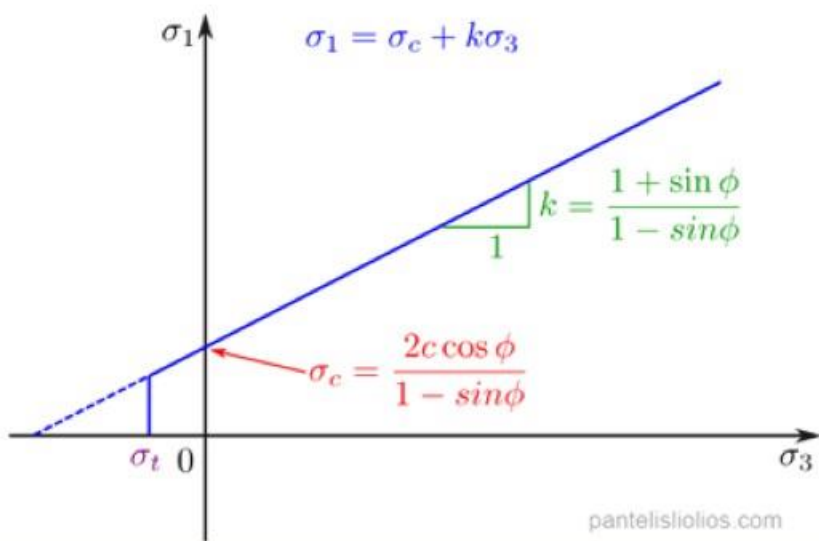
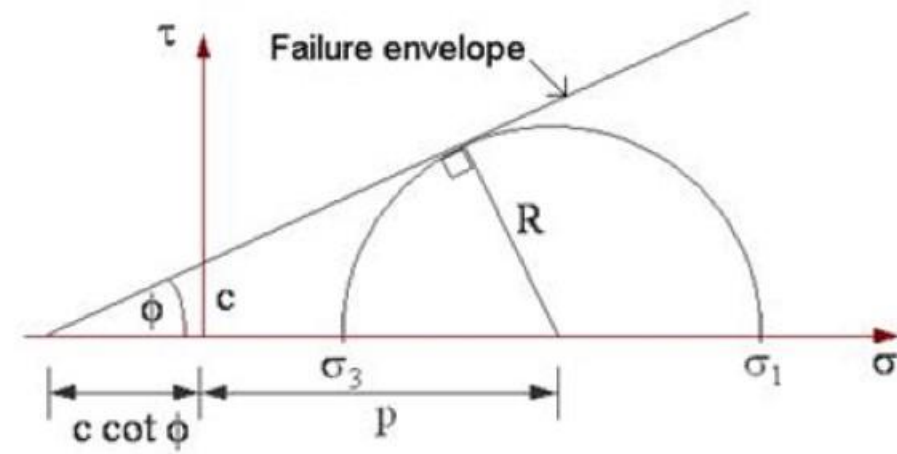
- ✓ Rearranging, $\sigma_1 = \sigma_3 \left(\frac{1 + \sin \varphi}{1 - \sin \varphi} \right) + 2c \sqrt{\frac{1 + \sin \varphi}{1 - \sin \varphi}}$

- ✓ Where, $\frac{1 + \sin \varphi}{1 - \sin \varphi} = \left(\frac{\cos \frac{\varphi}{2} + \sin \frac{\varphi}{2}}{\cos \frac{\varphi}{2} - \sin \frac{\varphi}{2}} \right)^2 = \tan^2 \left(\frac{\pi}{4} + \frac{\varphi}{2} \right)$



Mohr's Failure Criteria

- ✓ When uniaxial compression $\sigma_3 = 0$; $c c$
- ✓ When pure tension $\sigma_1 = 0$; $\sigma_t = 2c \sqrt{\frac{1-\sin \phi}{1+\sin \phi}} = \frac{2c \cos \phi}{1+\sin \phi}$
- ✓ $\frac{\sigma_c}{\sigma_t} = \frac{1+\sin \phi}{1-\sin \phi}$; therefore, $\sigma_1 = \sigma_3 \left(\frac{\sigma_c}{\sigma_t} \right) + \sigma_c$
- ✓ $2\beta = \frac{\pi}{2} + \phi$ or, $\beta = \frac{\pi}{4} + \frac{\phi}{2}$



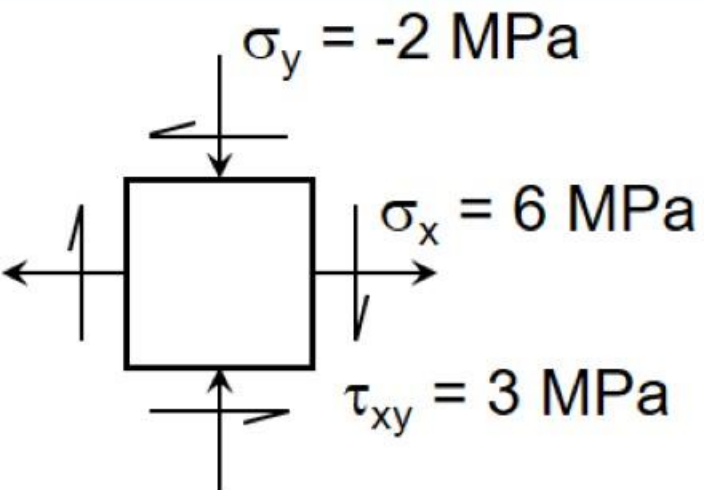


Construction of Mohr's Circle

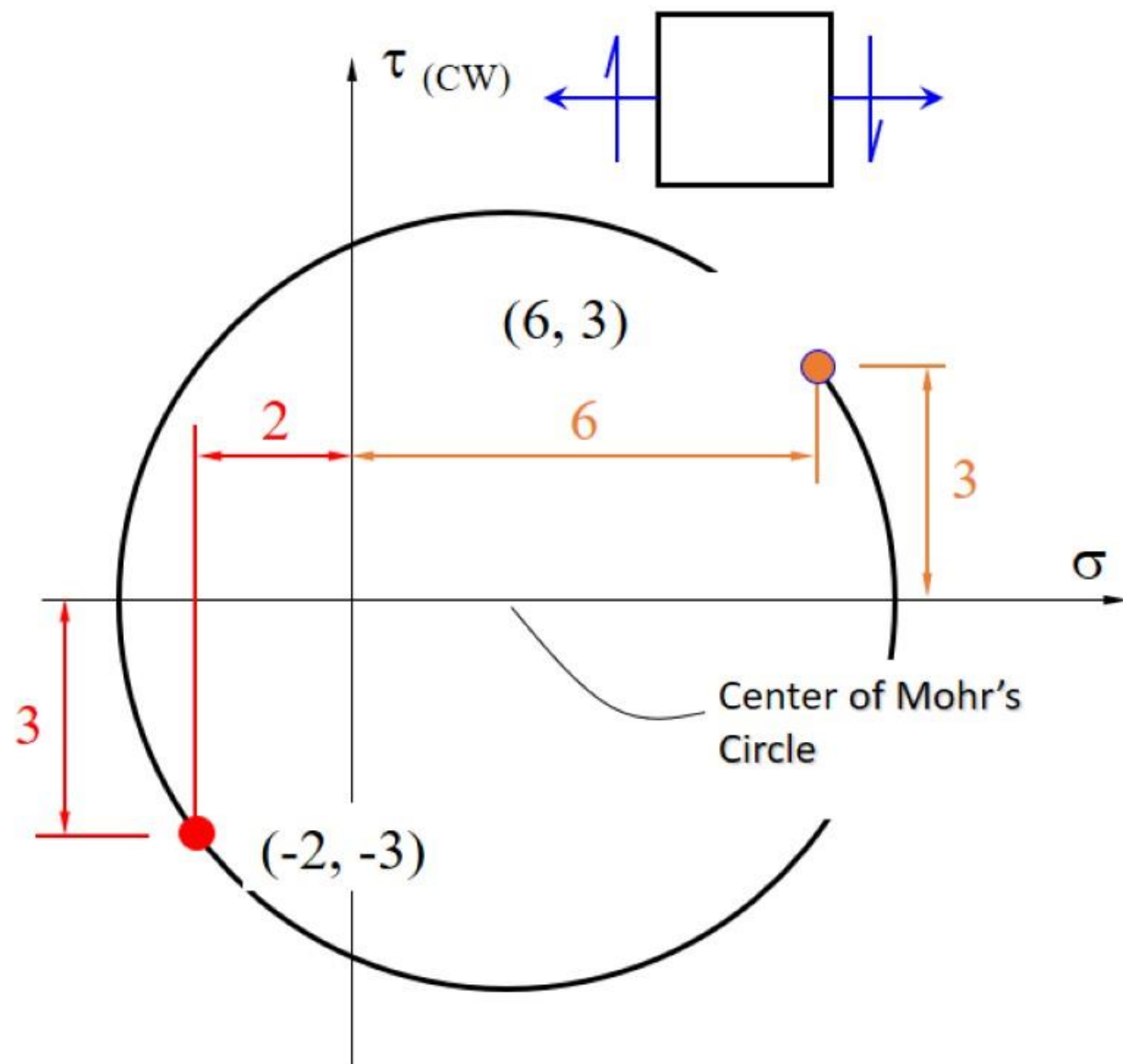
1. Determine the point on the body in which the principal stresses are to be determined.
2. Treating the load cases independently and calculated the stresses for the point chosen.
3. Choose a set of x-y reference axes and draw a square element centered on the axes.
4. Identify the stresses σ_x , σ_y , and $\tau_{xy} = \tau_{yx}$ and list them with the proper sign.
5. Draw a set of $\sigma - \tau$ coordinate axes with σ being positive to the right and τ being positive in the upward direction. Choose an appropriate scale for the each axis.
6. Using the rules on the previous page, plot the stresses on the x face of the element in this coordinate system (point V). Repeat the process for the y face (point H).
7. Draw a line between the two point V and H. The point where this line crosses the σ axis establishes the center of the circle.
8. Draw the complete circle.
9. The line from the center of the circle to point V identifies the x axis or reference axis for angle measurements (i.e. $\theta = 0$).

Note: The angle between the reference axis and the σ axis is equal to $2\theta_p$

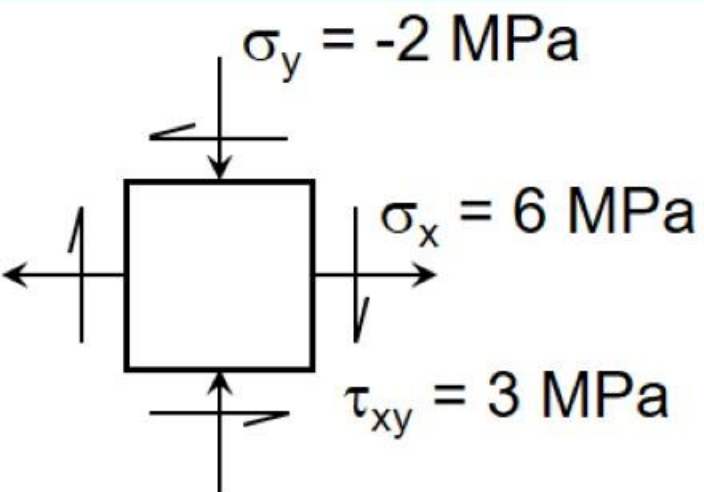
Exercise- Questions



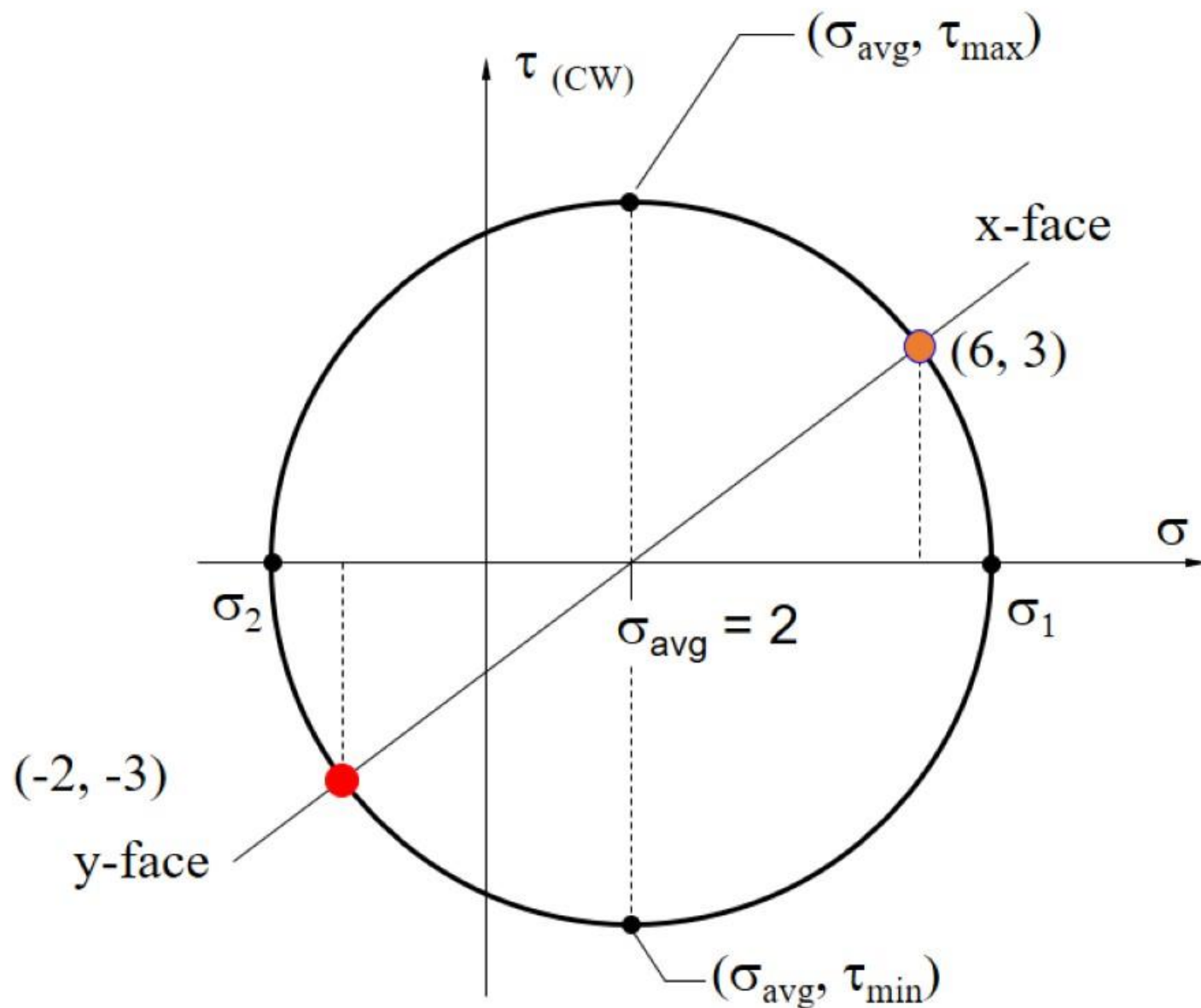
- ✓ Sign Convention for normal stress:
Tensile is positive and compression is negative
- ✓ Sign Convention for shear stress:
Clockwise is positive and counter-clockwise is negative



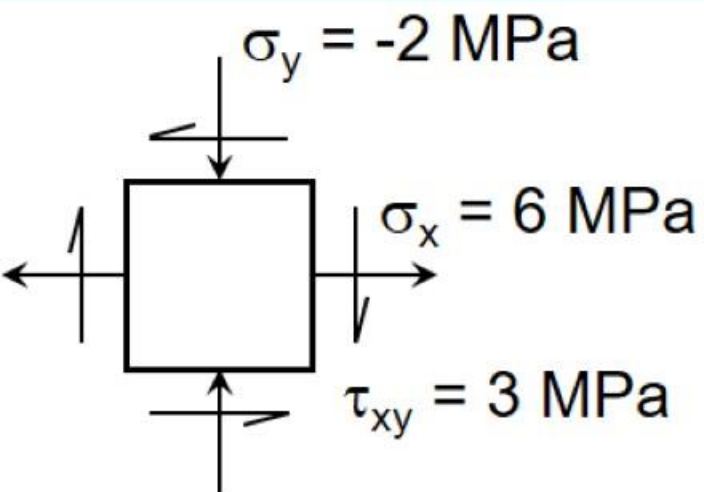
Exercise- Questions



$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 2 \text{ ksi}$$



Exercise- Questions



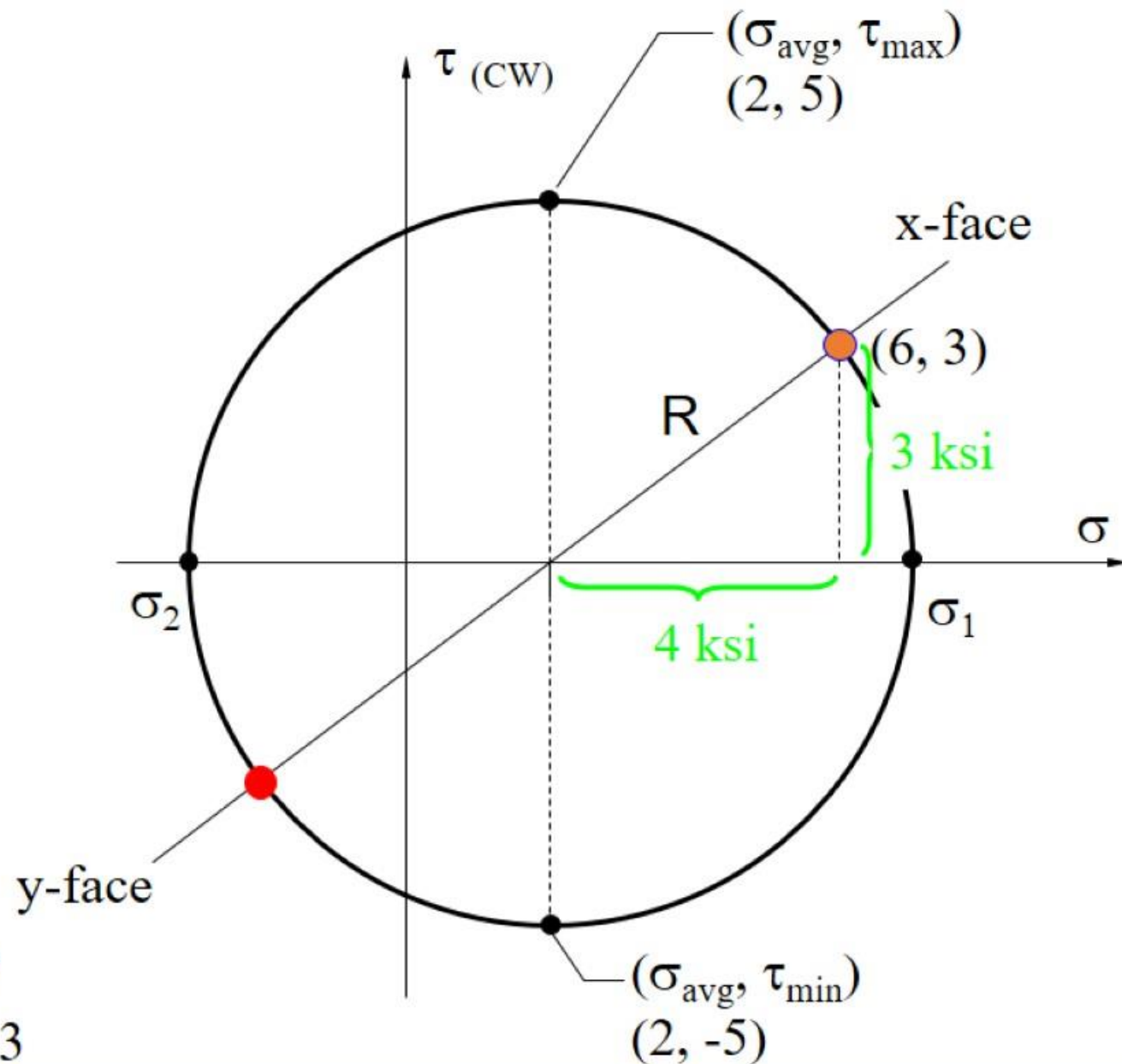
$$R = \sqrt{(3)^2 + (4)^2}$$

$$= 5$$

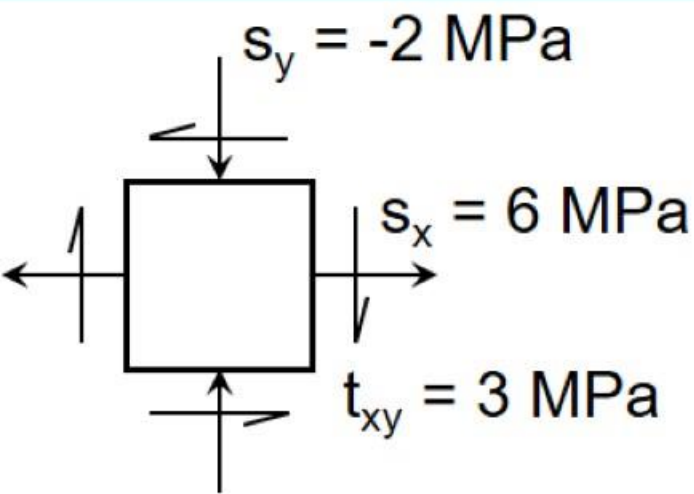
$$R = \tau_{\max}$$

$$\sigma_1 = \sigma_{\text{avg}} + R = 7$$

$$\sigma_2 = \sigma_{\text{avg}} - R = -3$$



Exercise- Questions

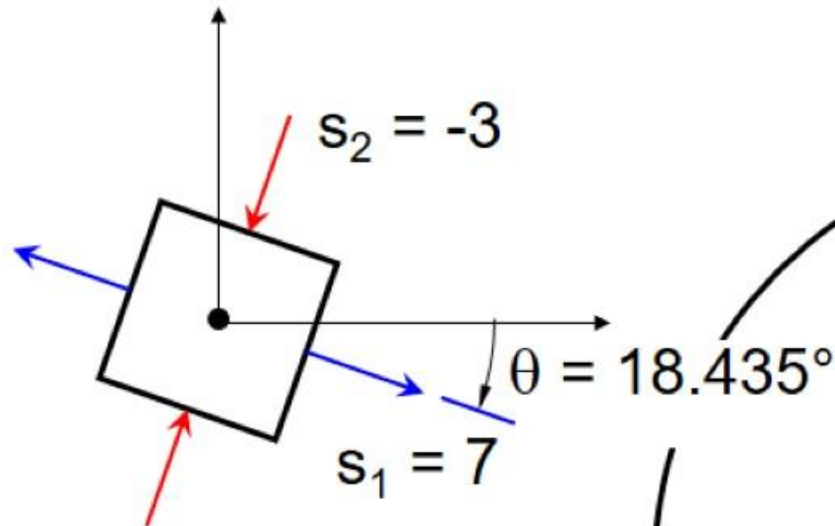


$$2\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$2\theta = 36.869^\circ$$

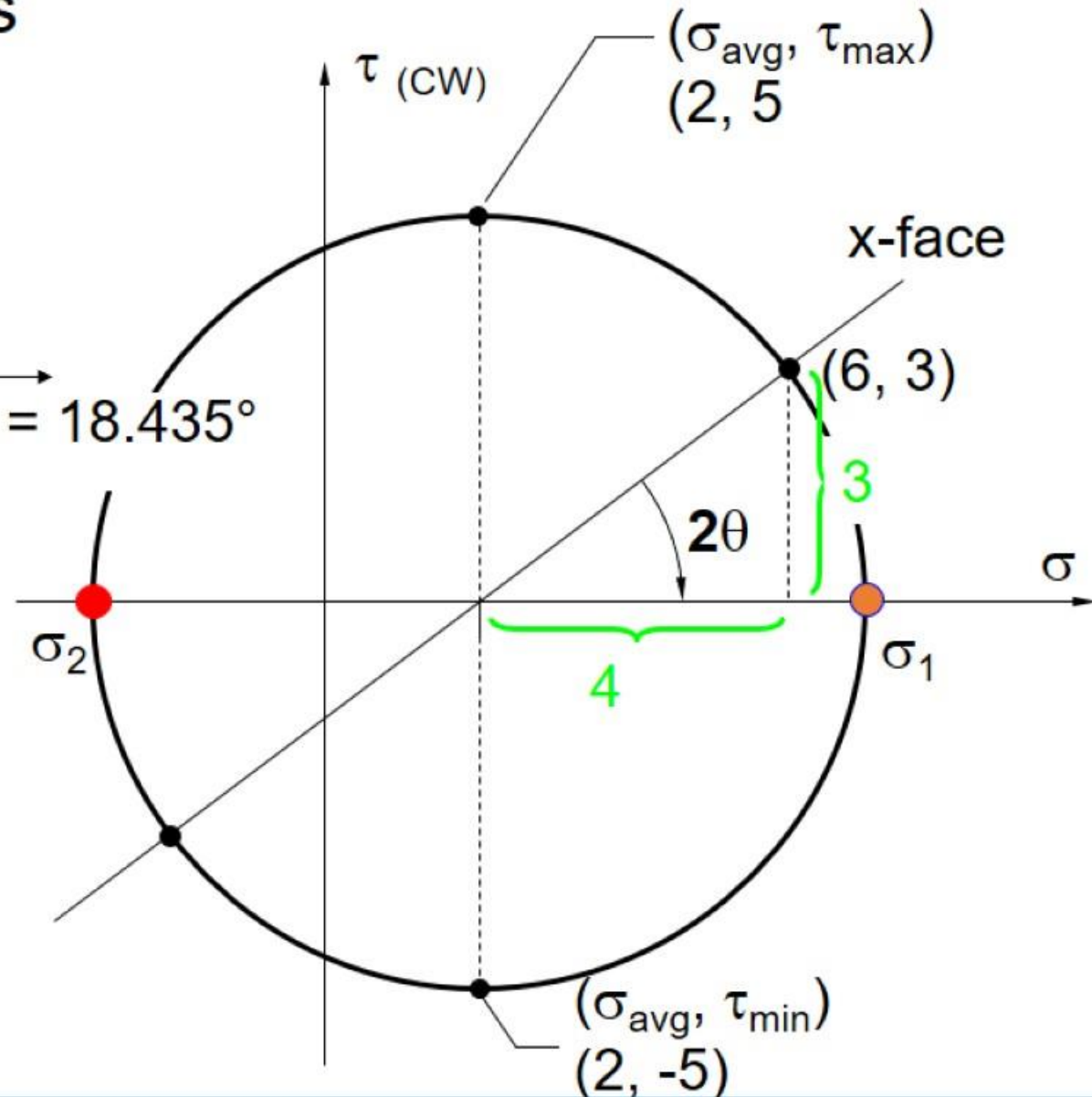
$$\theta = 18.435^\circ$$

Principle Stress

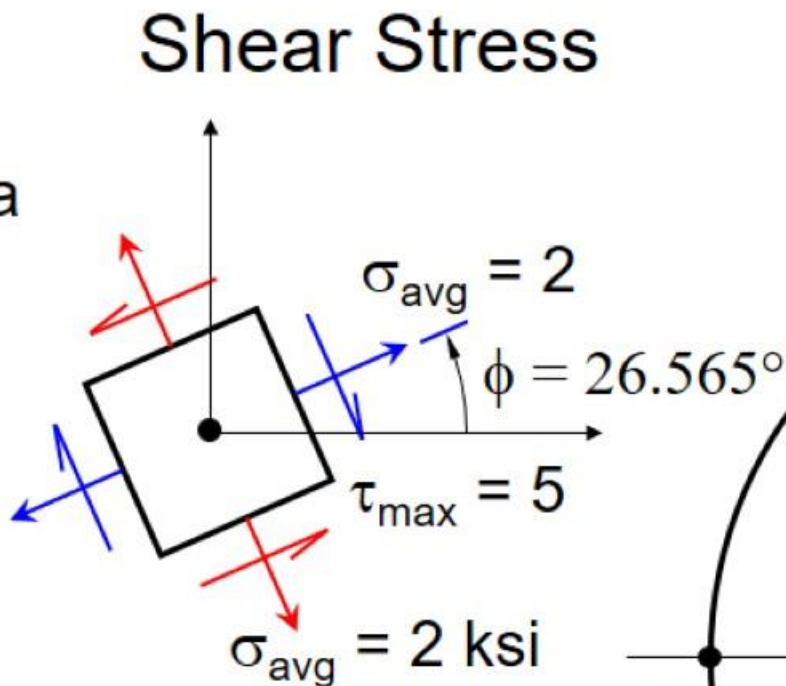
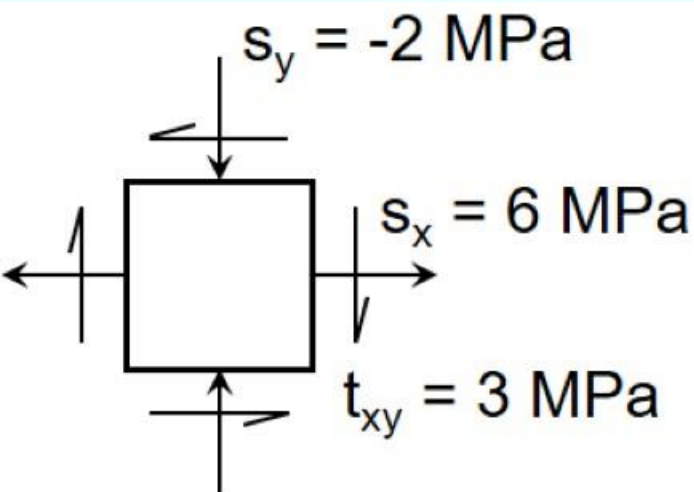


Principle Stress Element

Rotation on element is half of the rotation from the circle in same direction from x-axis



Exercise- Questions



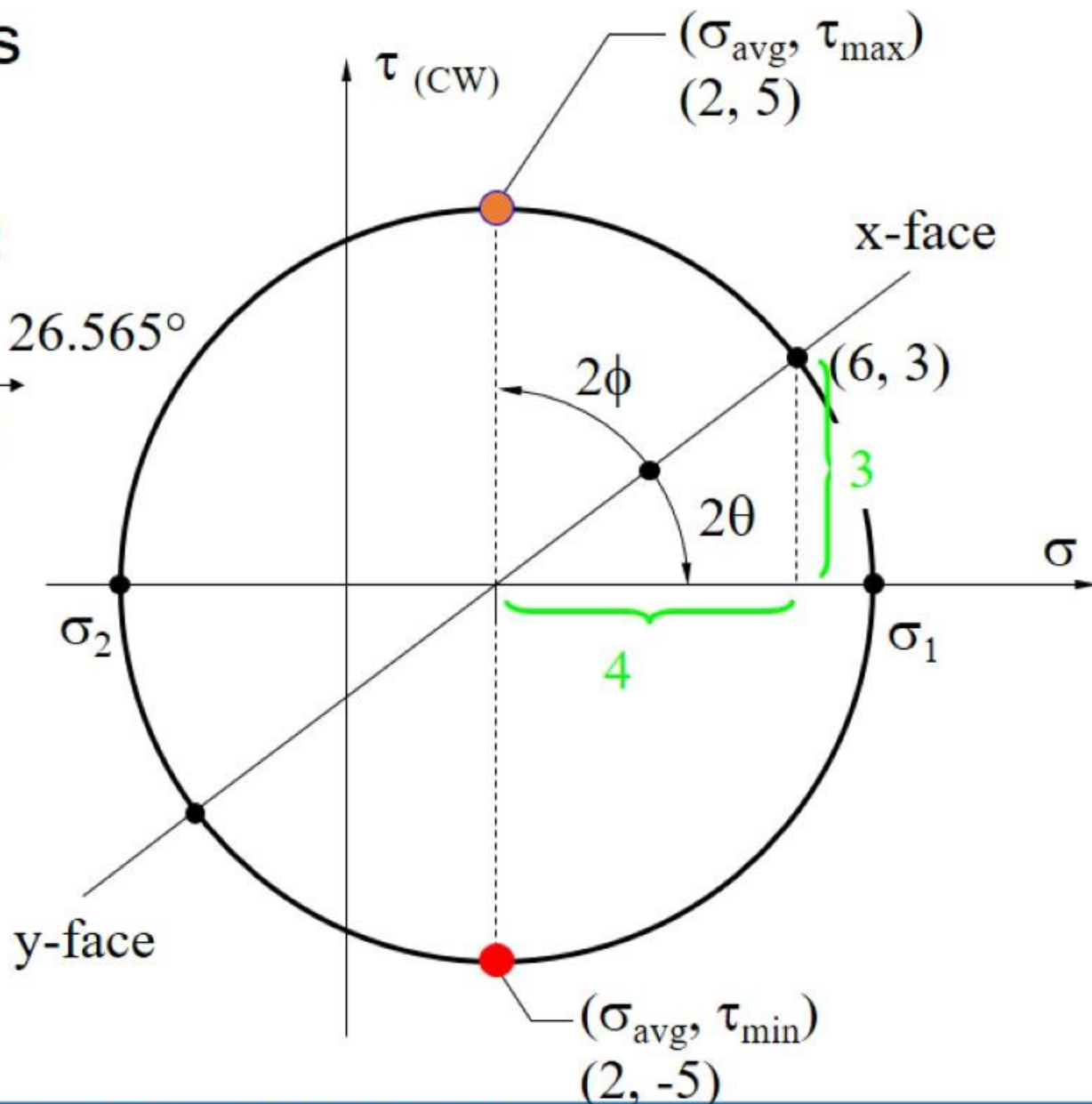
Maximum Shear Stress Element

$$2\phi = 90^\circ - 2\theta$$

$$2\phi = 90 - 36.869^\circ$$

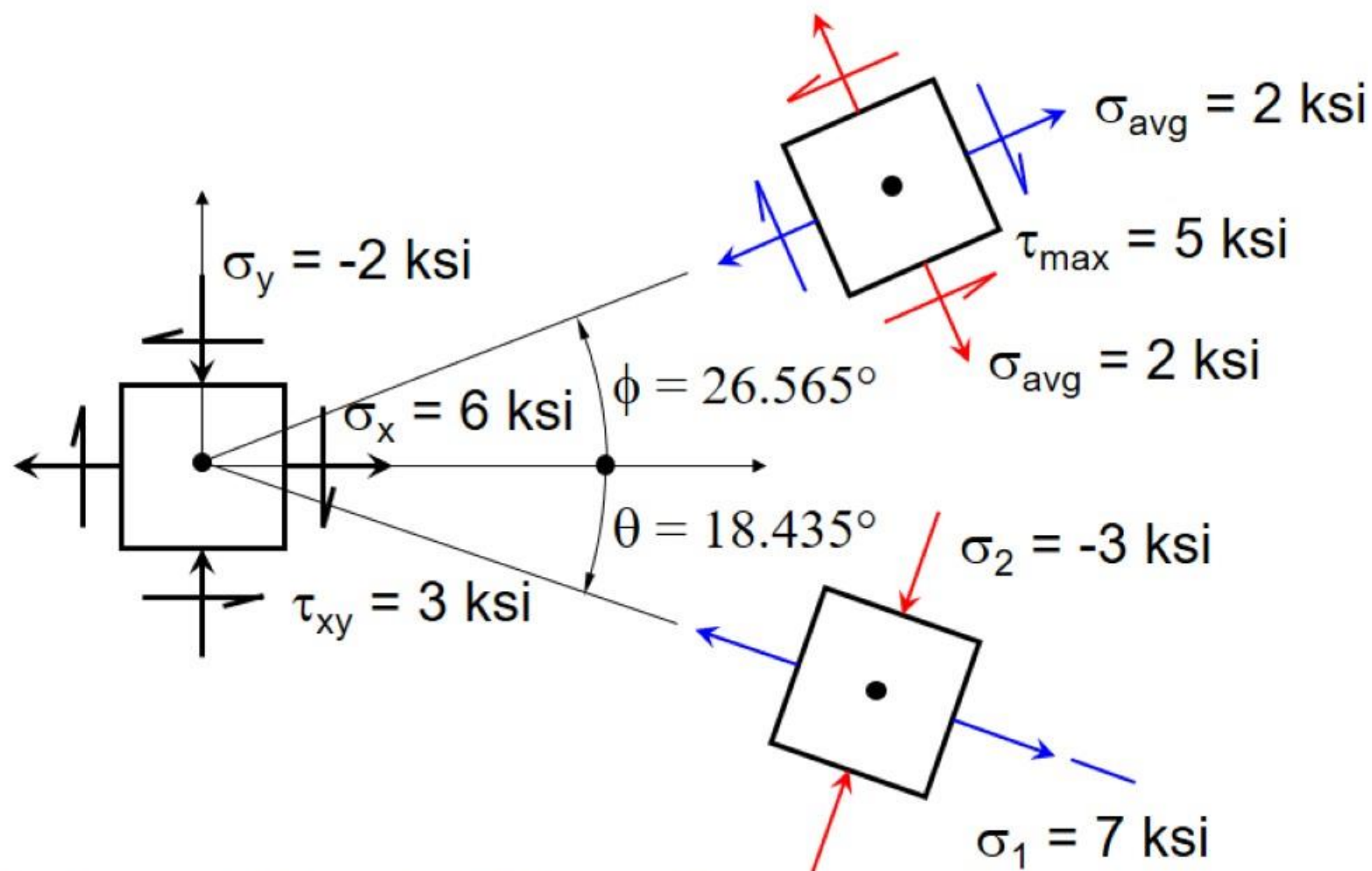
$$2\phi = 53.130^\circ$$

$$\phi = 26.565^\circ$$



Exercise- Questions

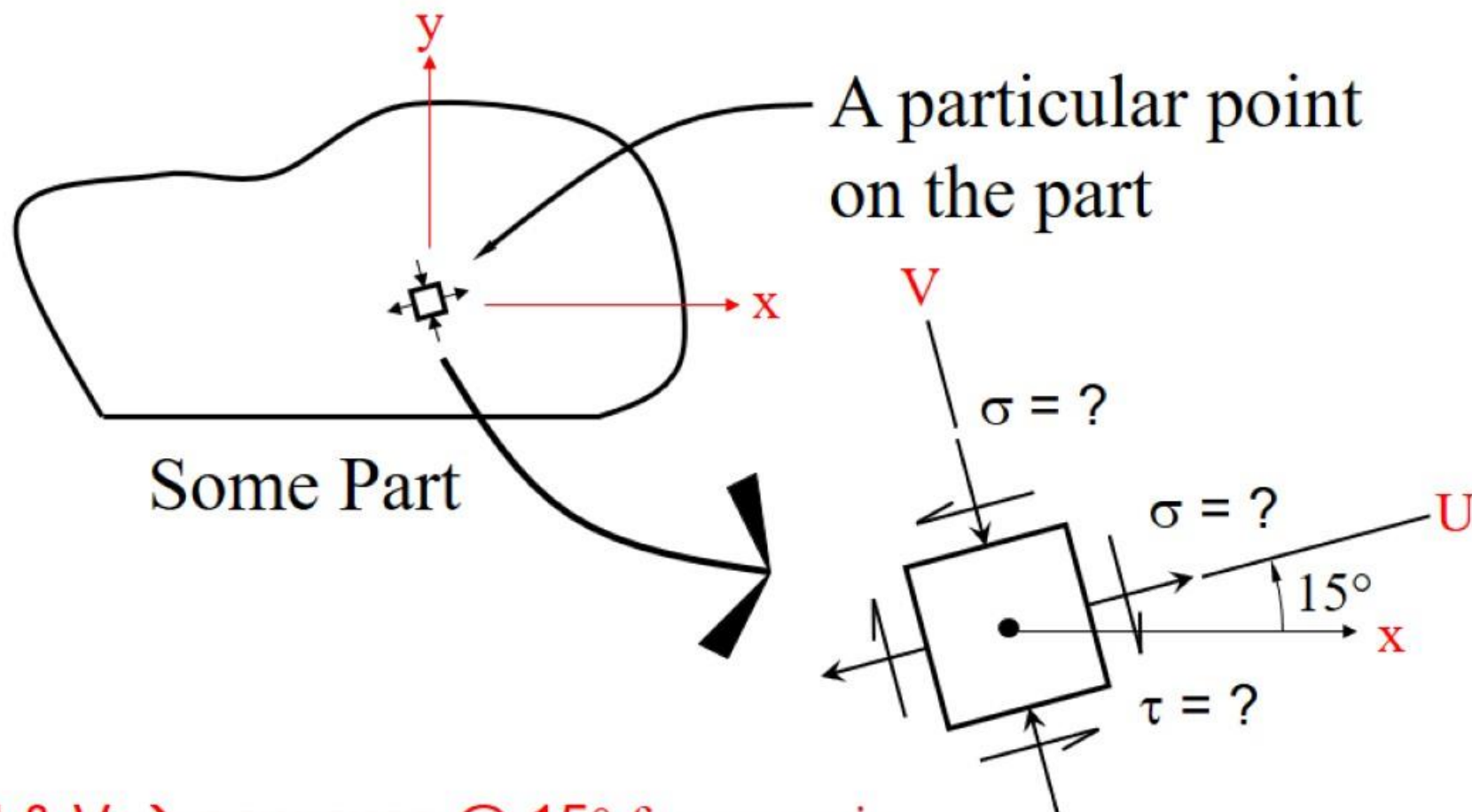
Relationship Between Elements



$$\theta + \phi = 18.435^\circ + 26.565^\circ = 45^\circ$$

Exercise- Questions

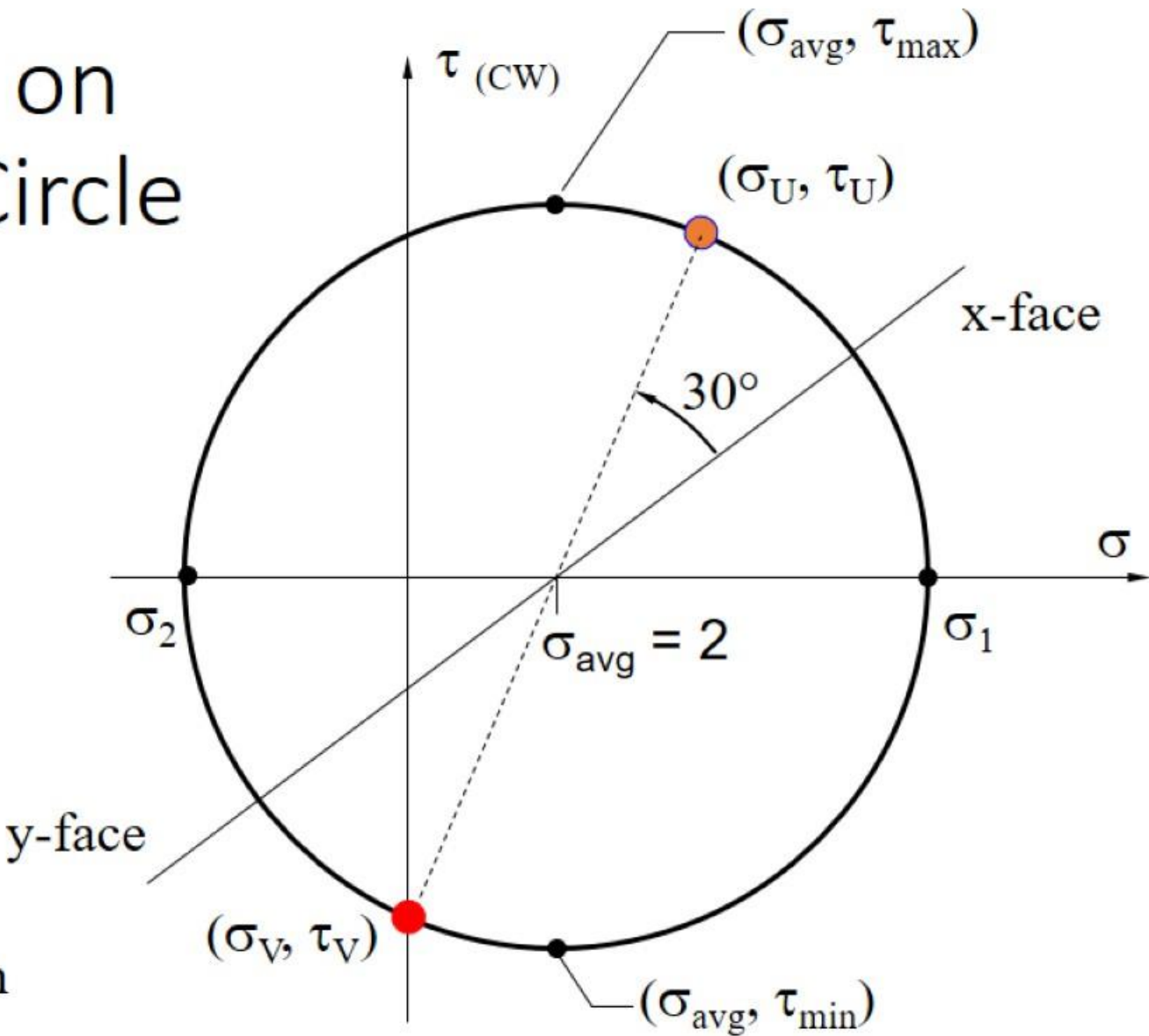
What's the stress at angle of 15° CCW from the x-axis?



U & V $\rightarrow$ new axes @ 15° from x-axis

Exercise- Questions

Rotation on Mohr's Circle



15° on part and element is 30° on Mohr's Circle

Exercise- Questions

Rotation on Mohr's Circle

$$\sigma_U = \sigma_{avg} + R \cdot \cos(66.869^\circ)$$

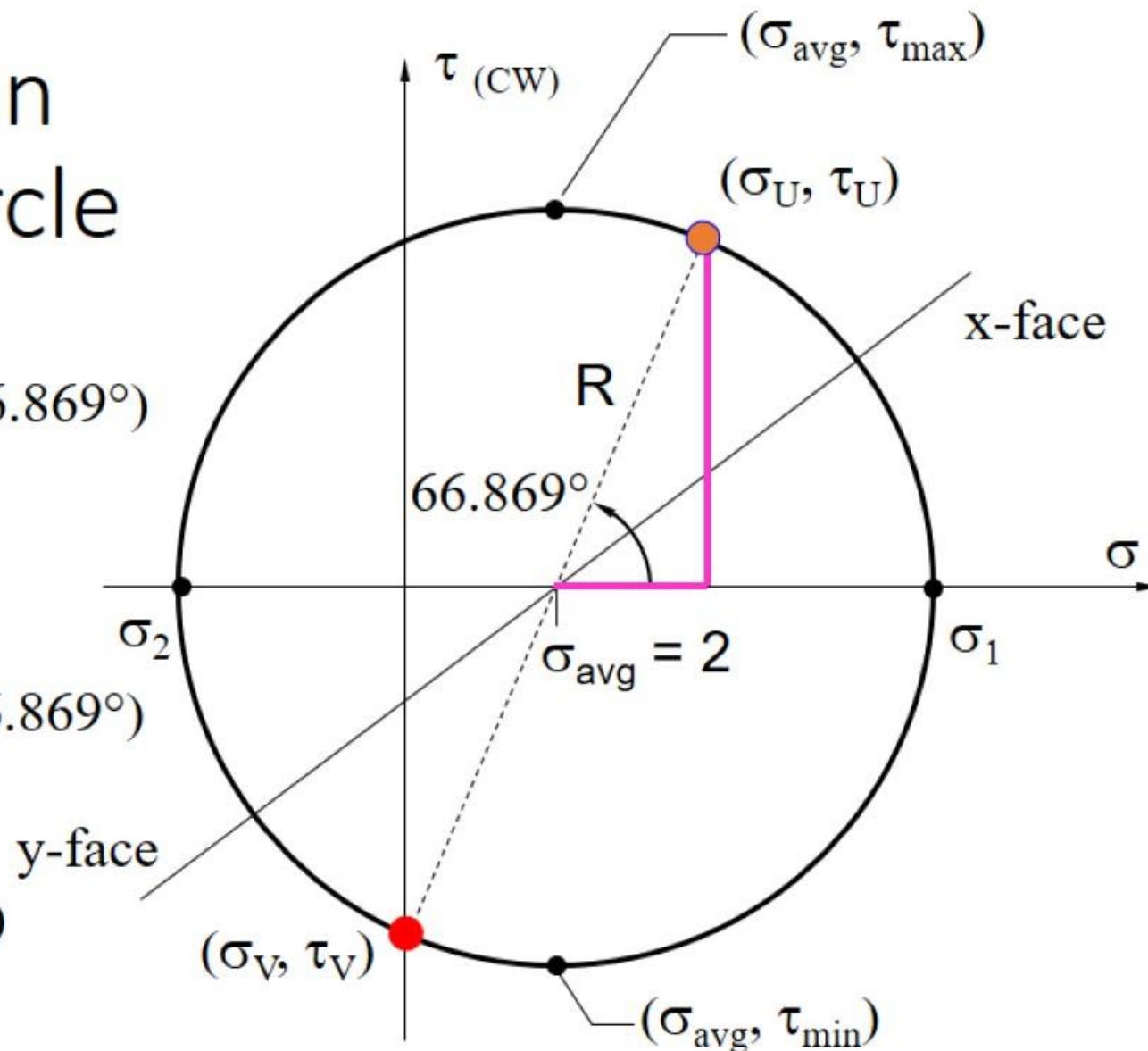
$$\sigma_U = 3.96$$

$$\sigma_V = \sigma_{avg} - R \cdot \cos(66.869^\circ)$$

$$\sigma_V = 0.036$$

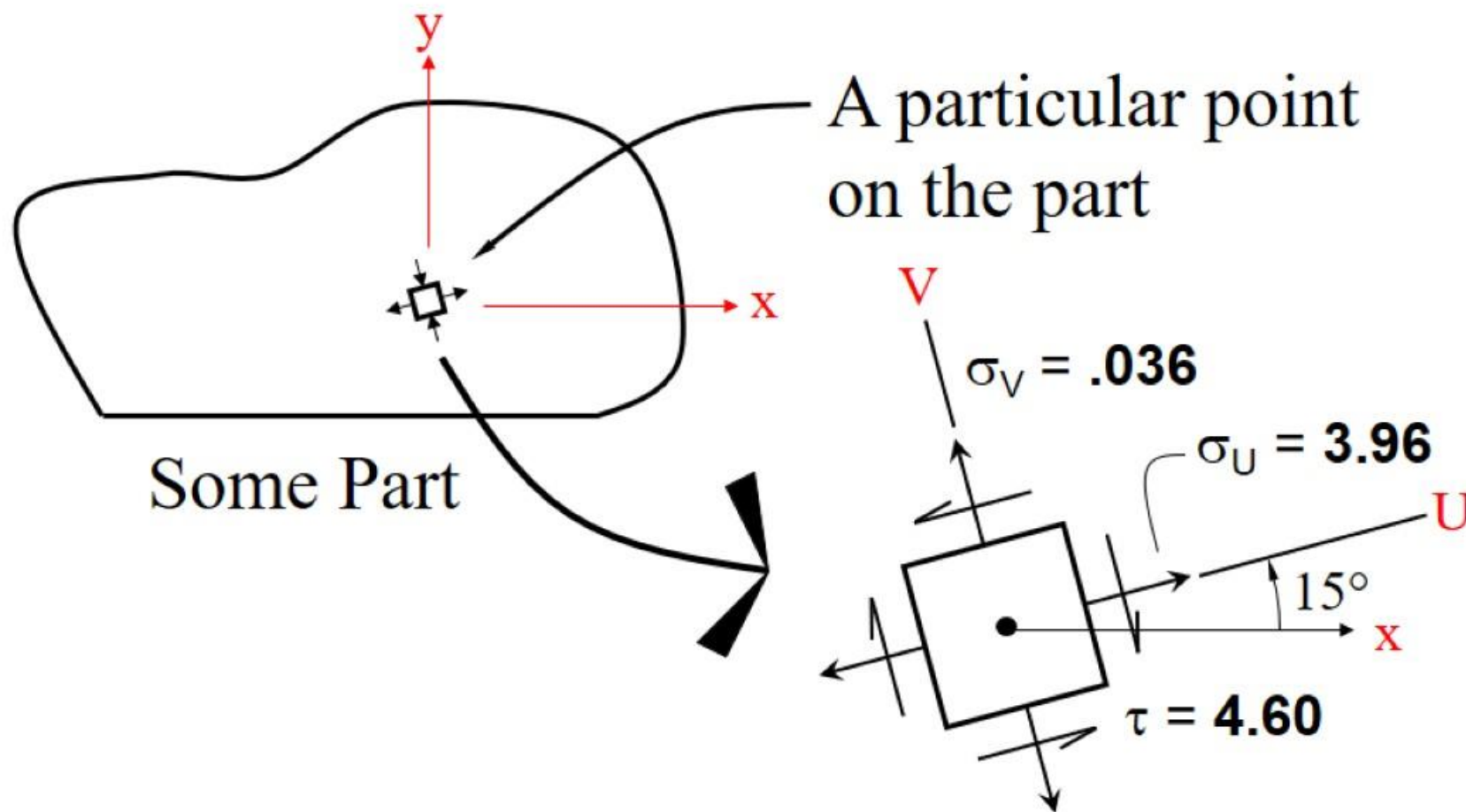
$$\tau_{UV} = R \cdot \sin(66.869^\circ)$$

$$\tau_{UV} = 4.60$$



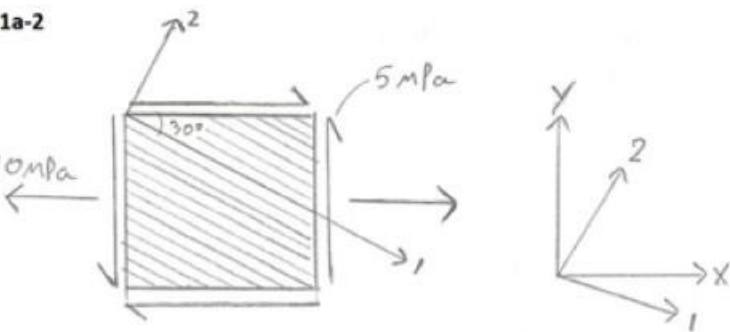
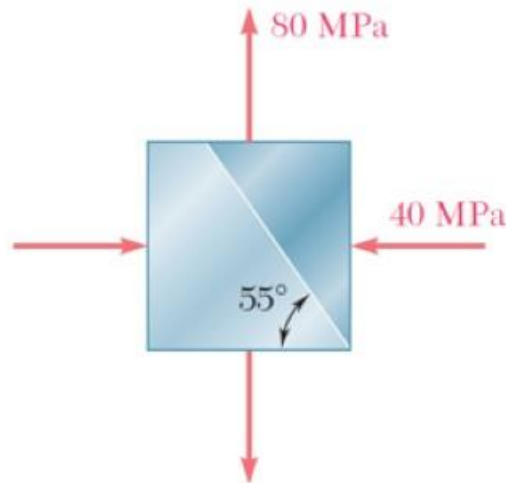
Exercise- Questions

What's the stress at angle of 15° CCW from the x-axis?



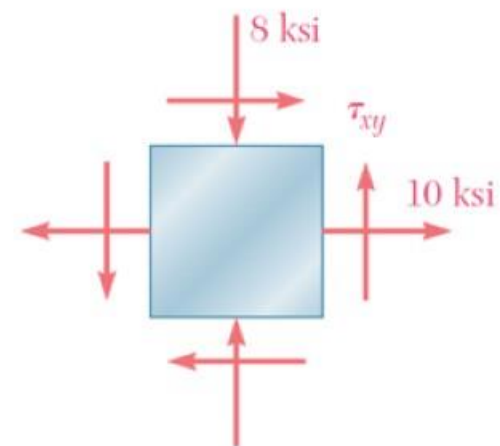
Exercise- Questions

21a-1 For the given state of stress, determine the normal and shearing stresses exerted on the oblique face of the shaded triangular element shown. Use a method of analysis based on the equilibrium of that element, as was done in the derivations of Sec. 7.2.

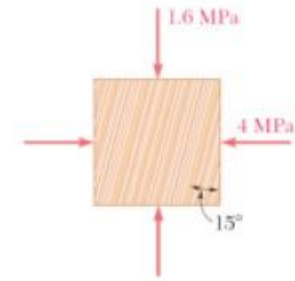


A test sample for a unidirectional carbon fiber laminate is subjected to the combined loading shown. Determine the shear stress parallel to the fibers (along the "1" axis), and the normal stress perpendicular to the fibers (along the "2" axis).

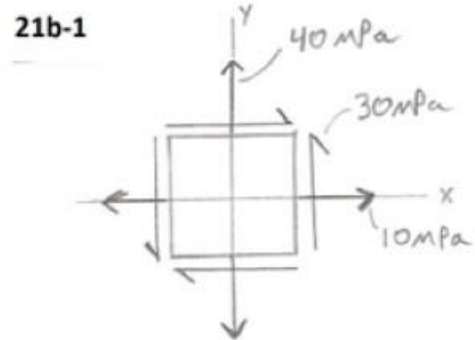
21a-3 For the state of plane stress shown, determine (a) the largest value of τ_{xy} for which the maximum in-plane shearing stress is equal to or less than 12 ksi, (b) the corresponding principal stresses.



21b-3 The grain of a wooden member forms an angle of 15° with the vertical. For the state of stress shown, determine (a) the in-plane shearing stress parallel to the grain, (b) the normal stress perpendicular to the grain.

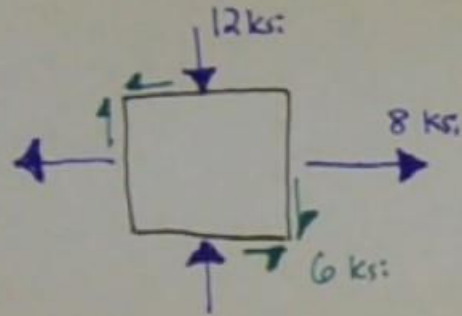


Solve using Mohr's circle (make sure you construct and label a detailed diagram of the circle)



For the loading on the element shown:
a) Construct Mohr's circle (be neat!)
b) Use this circle to calculate/find the principal stresses of the element
c) Use the circle to determine the maximum shearing stress present in the element

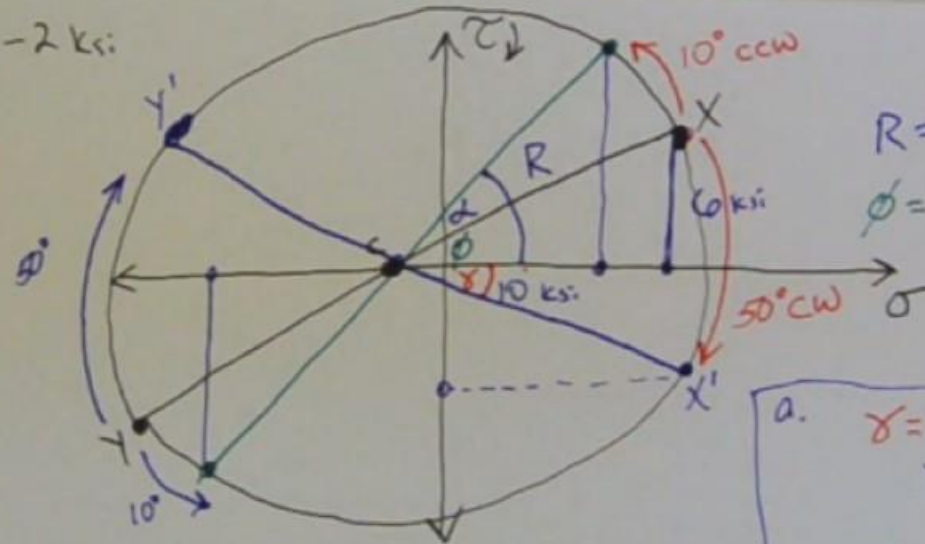
Exercise- Questions



For the given state of stress, determine the normal & shearing stress after the element has been rotated:

- 25° CW
- 10° CCW

$$\left. \begin{aligned} \sigma_x &= 8 \text{ ksi} \\ \sigma_y &= -12 \text{ ksi} \\ \tau_{xy} &= -6 \text{ ksi} \end{aligned} \right\} \sigma_{avg} = \frac{\sigma_x + \sigma_y}{2} = -2 \text{ ksi}$$



$$R = \sqrt{10^2 + 6^2} = 11.66 \text{ ksi}$$

$$\phi = 2\theta$$

$$\tan 2\theta = \frac{6}{10}$$

$$\phi = 2\theta = 30.96^\circ$$

b. $\alpha = 30.96 + 20 = 50.96^\circ$

$$\sigma_{x'} = 5.34 \text{ ksi}$$

$$\sigma_{y'} = -9.34 \text{ ksi}$$

$$\tau_{x'y'} = -9.06 \text{ ksi}$$

a. $\gamma = 50^\circ - 30.96^\circ$
 $\gamma = 19.04^\circ$

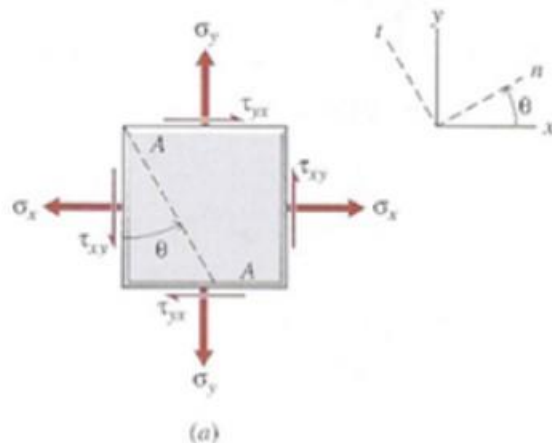
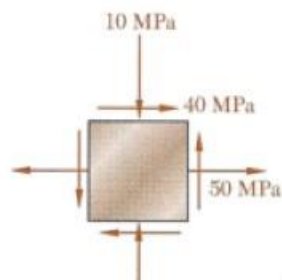
$$\sigma_{x'} = \sigma_{avg} + R \cos \gamma = 9.02 \text{ ksi}$$

$$\sigma_{y'} = \sigma_{avg} - R \cos \gamma = -13.02 \text{ ksi}$$

$$\tau_{x'y'} = R \sin \gamma = 3.80 \text{ ksi}$$

Exercise- Questions

Example Problem:



Determine:

- The principal planes.
- The principal stresses.
- The maximum shear stress and the corresponding normal stress.

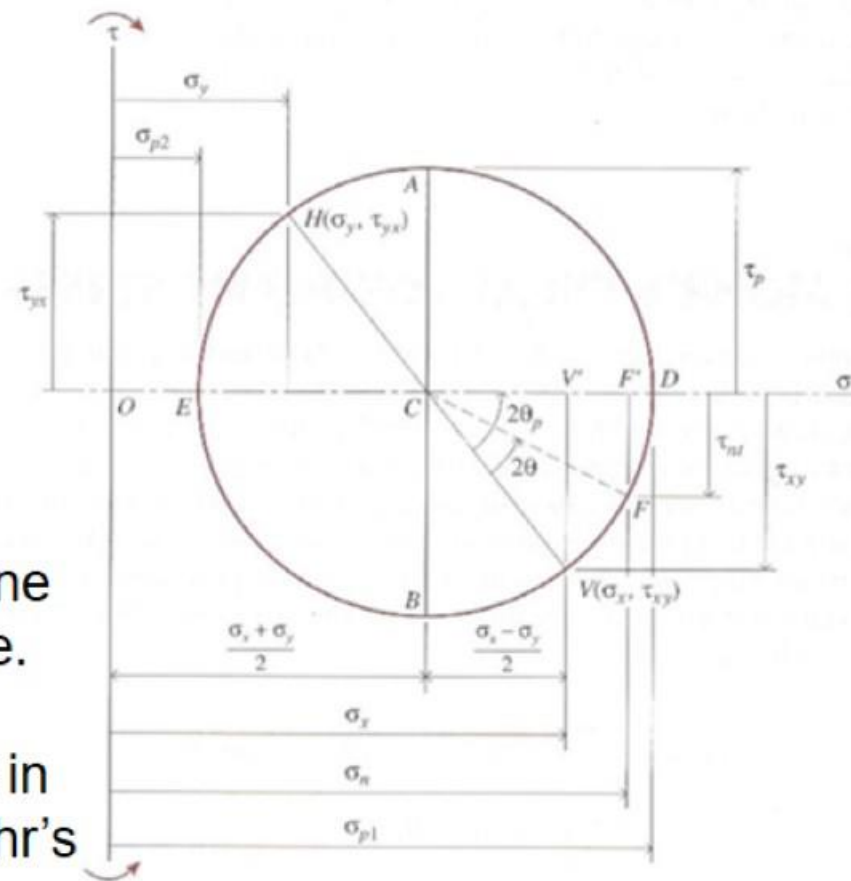
Mohr's Circle for Plane Stress: The transformation equations for plane stress can be represented in a graphical format known as Mohr's circle. This representation is useful in visualizing the relationships between normal and shear stresses acting on various inclined planes at a point in a stressed body. Before we discuss the procedure for constructing Mohr's circle there are several rules that apply.

Stress Component

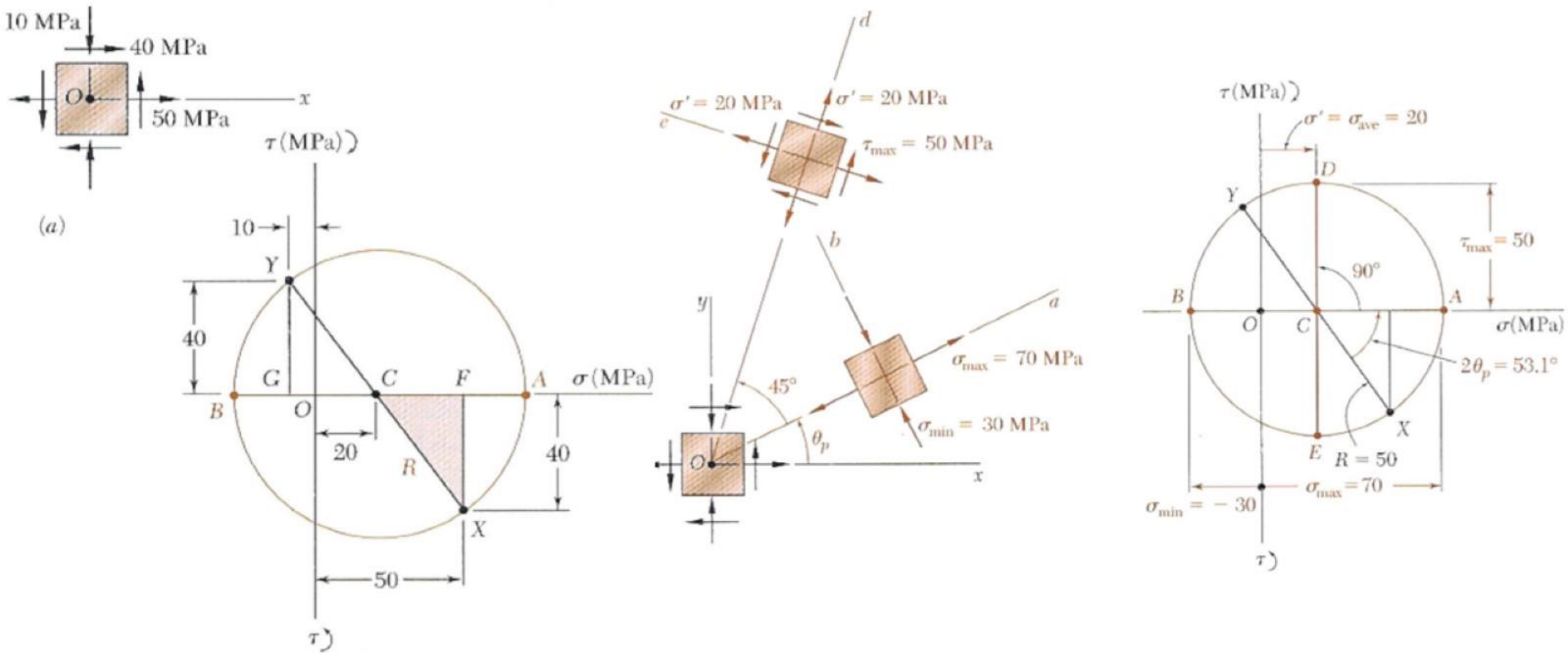
Plot Normal Stresses + For Tension (Horizontal Axis) - For Compression

Shear Stresses

+ For Clockwise Shear (Vertical Axis) - For Counterclockwise Shear



Exercise- Questions



Thank You

Queries??

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